LINA - Lab05 – Vectors(2D)

Work individually and make sure that throughout the course of this lab, you re-save sufficiently your solutions file on your own machine, finally delivered .**PDF**

LINA_2627_IGP99E_Van_Der_Zever_Johnny_Lab**05**.**PDF** (< 5 MB),

Reduce via https://www.ilovepdf.com/compress_pdf or Adobe Acrobat (Online)

➔ which is to submit via week-folder Leho/ LINA / Week**05**/Lab**05** Practice

Also, make sure to repeat Class group + Full Name +Lab**05** number inside it.

1. Exercises

1.1. Basic exercises

Make these exercises on paper and then verify your results in GeoGebra.

GeoGebra tips:

- Creating a vector (Input bar/Vector command)

  Input: Vector(<Start Point>, <End Point>)
- Drawing a vector (Graphics toolbar/Vector button)

  
- Length of a vector (Input bar/Length command)

  Input: Length(<Object>)
- Components of a vector (automatically in Algebra view)

  Vector

  $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

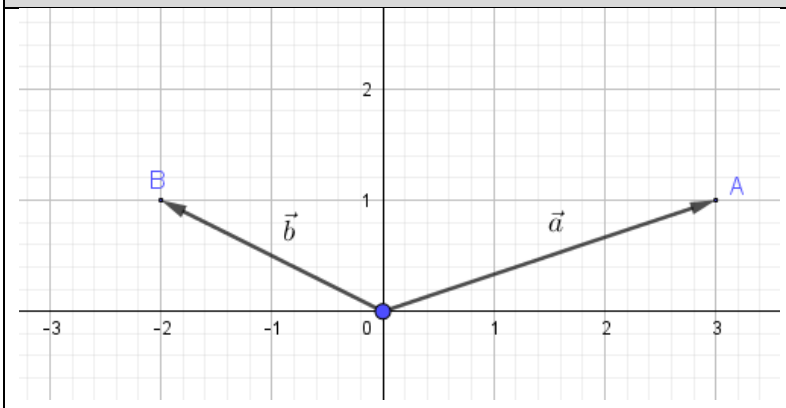
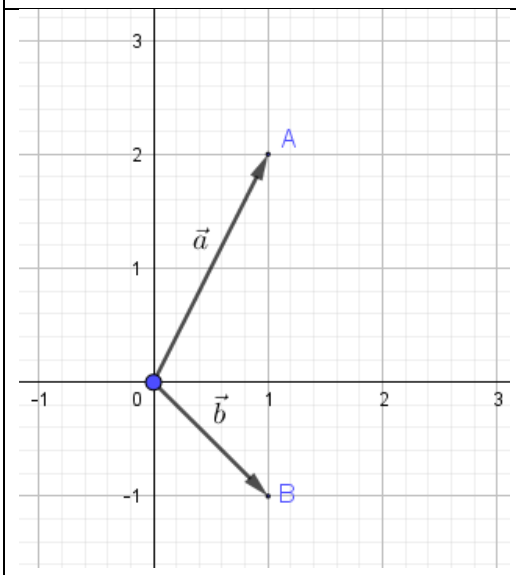
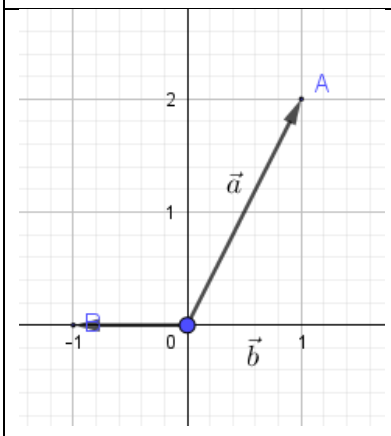
1) In GeoGebra:
   - Create a slider with the name **scale**
   - Define 3 vectors in the input bar, like this:
     $\vec{a}$: scale * Vector((6, 2))
     $\vec{b}$: scale * Vector((8,4))
     $\vec{c}$: scale * Vector((2,4))
   - Select the Polygon button  and draw a triangle that connects the 3 endpoints of these vectors.
   - Then change the scale value using the slider, what do you notice?

1.1.1. Subtraction

The subtraction of $\vec{a}$ and $\vec{b}$ can be written as the addition of $\vec{a}$ and $-\vec{b}$

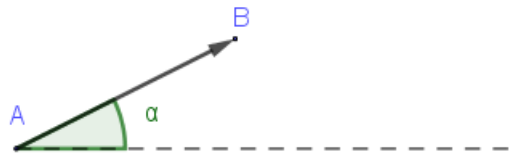
$\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$

Subtract the vectors $\vec{a}$ and $\vec{b}$ in 2 ways.

Head-to-tail method	Subtracting components
	
	
	

1.2. Bridging exercises

1.2.1. Decompose a 2D vector trigonometrically



Give the formulas for the x and y components of the vector $\overrightarrow{AB}$ when the angle α with the positive x-axis and the length l are given.

x =
y =

Apply these formulas. Calculate the components of the vectors for the given lengths and angles.

Vector	Length	Angle	x component	y component
$\vec{a}$	1	0°		
$\vec{b}$	2	30°		
$\vec{c}$	3	45°		
$\vec{d}$	4	90°		

1.2.2. Orthogonal vector

Calculate the x and y components of the vectors that are orthogonal to and have the same length as the vectors in previous exercise.

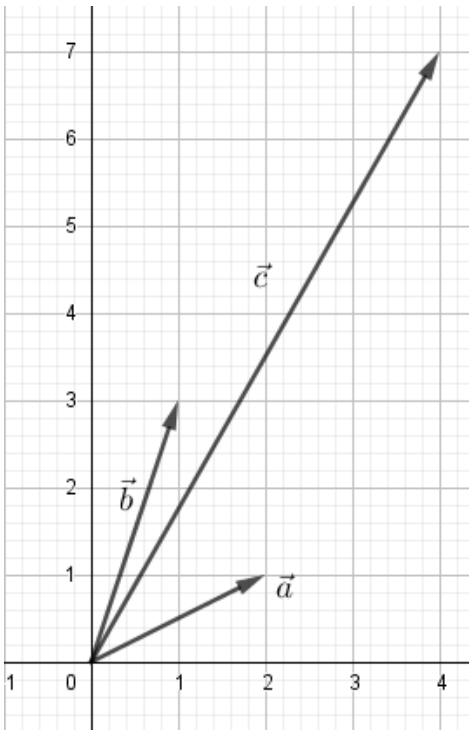
Given vector			Orthogonal vector		
Vector	Length	Angle	Angle	x component	y component
$\vec{a}$	1	0°	90°		
$\vec{b}$	2	30°	120°		
$\vec{c}$	3	45°	135°		
$\vec{d}$	4	90°	180°		

What can you conclude in general?

Try to write a general formula to find the orthogonal vector to a vector given in 2D space, when you would need the orthogonal vector of the same length as the given vector.

1.2.3. A vector as linear combination of 2 other vectors

Given are the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.



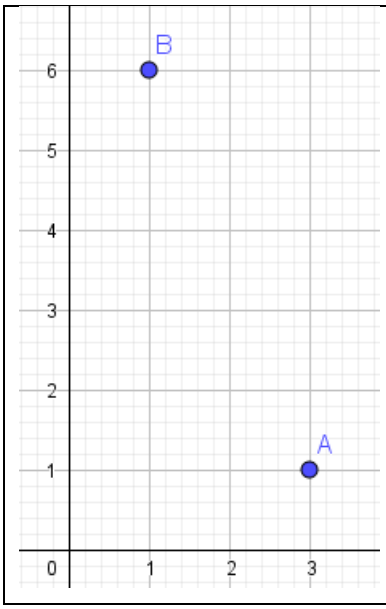
You can write the vector $\vec{c}$ as a linear combination of the vectors $\vec{a}$ and $\vec{b}$, like this

$\vec{c} = s * \vec{a} + t * \vec{b}$

Compute the values of s and t for this.

Verify your result in GeoGebra.

1.2.4. Distance between 2 points



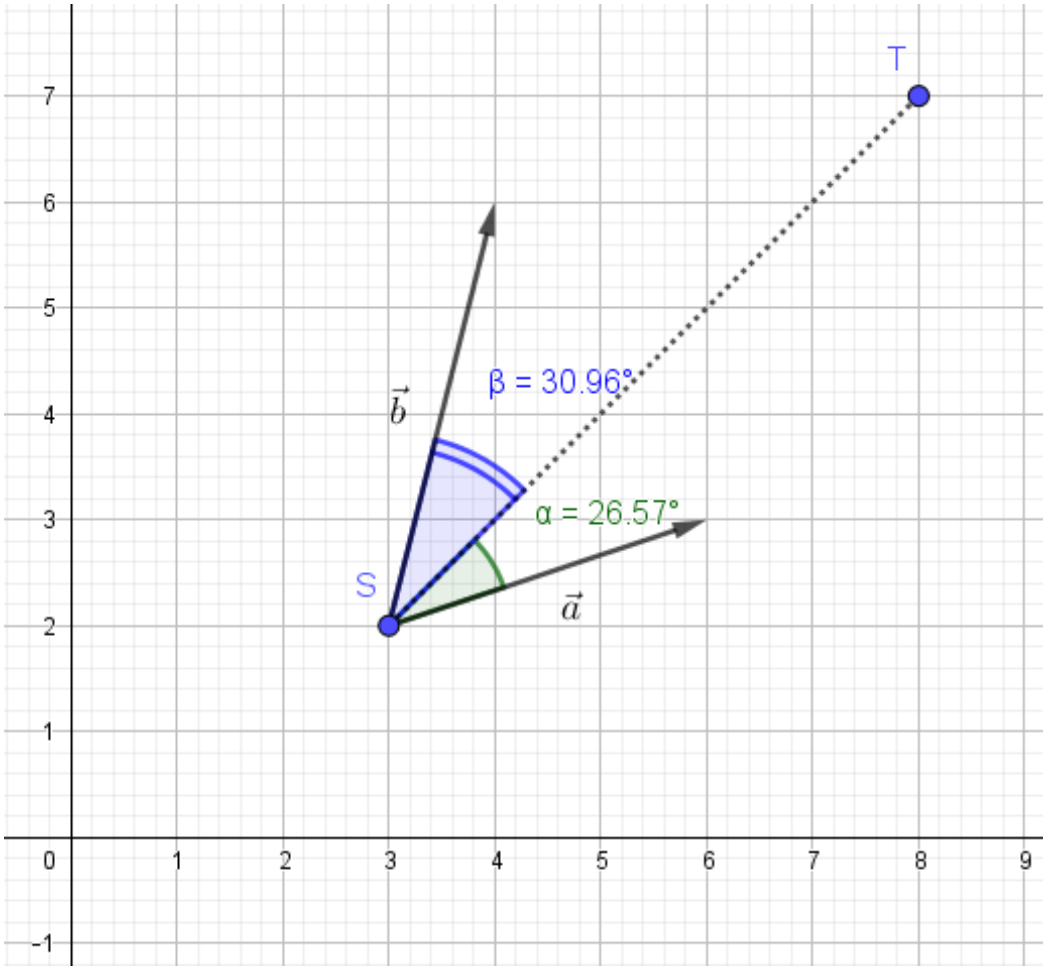
Given are 2 points A(3,1) and B(1,6). Calculate the distance between those 2 points (tip use vector subtraction and length of resulting vector)

1.3. Contextual practice

1.3.1. Vectors as forces

a. Pulling a box towards a target

Two teams are pulling a box located in a point S aiming for some target point approximately near T, the vectors $\vec{a}$ and $\vec{b}$ are the forces applied by team A and B respectively.



Calculate the resulting force vector applied on the box in S. (Hint: first determine the components of the forces $\vec{a}$ and $\vec{b}$)

How much force (magnitude) does each team apply on the box in S?

Answer here how much of each of the team's forces $\vec{a}$ and $\vec{b}$ is contributing into the direction of the target (which lies not exactly in portrayed point T).

b. Which team wins?

Three teams A, B and C are trying to pull a treasure into their own area.

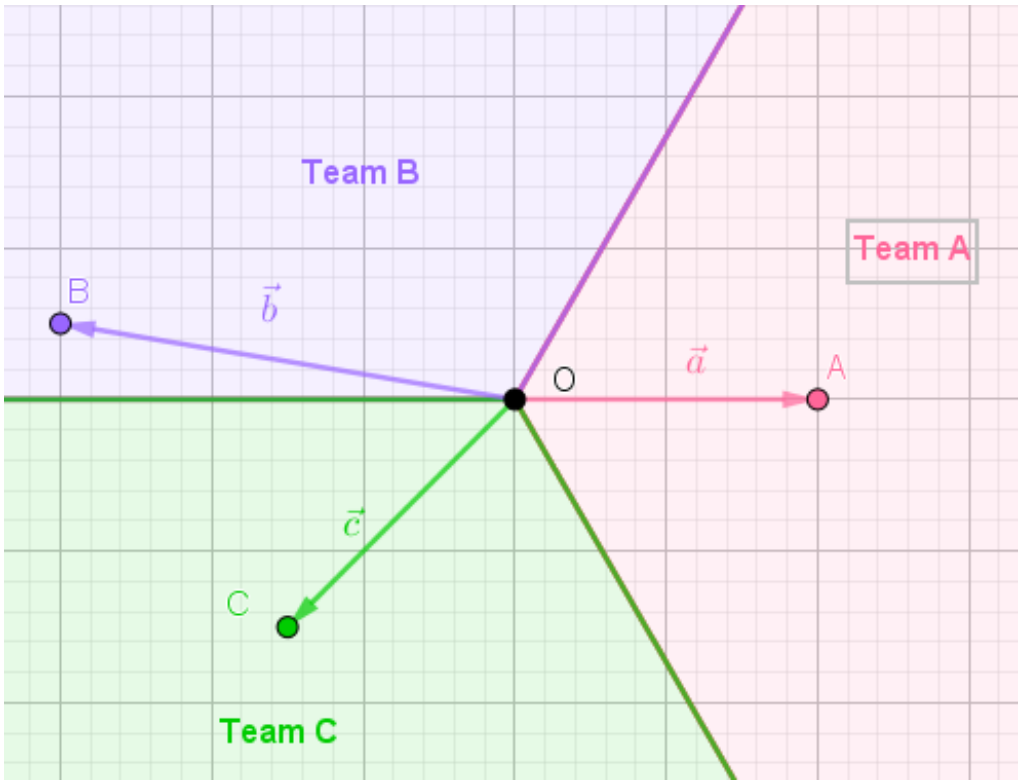
The treasure is initially located in the point O(4,2)

The forces (vectors) of each team start in point O, the end points are:

End point of force $\vec{a}$ of team A: (8, 2)

End point of force $\vec{b}$ of team B : (-2, 3)

End point of force $\vec{c}$ of team C : (1, -1)



Give the components of the location vectors corresponding with $\vec{a}$, $\vec{b}$ and $\vec{c}$

What is the magnitude of the resulting force $\vec{r}$ acting on the treasure?

Draw the resulting force that starts in point O.

To which camp is the treasure getting pulled?

1.3.2. Vectors as displacements

In a top-down game the hero moves through the level looking for some extra armour. He starts at a location S with coordinates (2, 1). The displacements (distance with direction) of the hero through the level are described by the following vectors:

$\vec{d}_1 = \begin{pmatrix} -5 \\ 1 \end{pmatrix}$

$\vec{d}_2 = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$

$\vec{d}_3 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$

$\vec{d}_4 = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$

Compute coordinates of the end point E he reached at the end of this expedition.

Compute the total length of the travelled distance.

At what distance d is the hero from its starting location?

1.3.3. Point in triangle

Have a look at the given file *LINA – Lab05 Vectors 1_3_3_PointInTriangle.ggb*
The vectors $\vec{a} = \binom{4}{2}$ and $\vec{b} = \binom{1}{4}$ coincide with two sides of triangle *tri*. Point P of the vector $\|\overrightarrow{OP}\|$ is defined as a linear combination of these 2 triangle vectors, like this $\binom{4*s + 1* t}{2*s + 4*t}$. The values s and t are controlled by 2 sliders. Move the point P by sliders and indicate (3rd column) locations of P with respect to the triangle ABC.

s	t	P
0	0	
1	0	
0	1	
Any value in [0,1]	0	
0	Any value in [0,1]	
s and t are positive and sum is 1		
At least one is negative		
Both are positive and the sum is < 1		

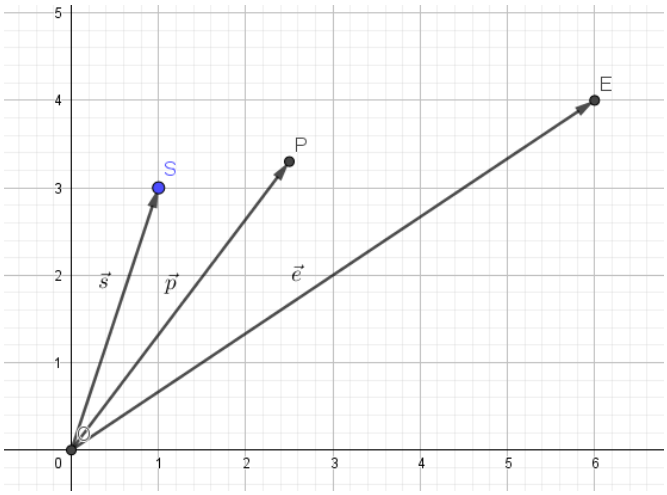
Some video is available via Point in triangle
https://www.youtube.com/watch?v=HYAgJN3x4GA

1.3.4. Linear interpolation between two vectors (lerp)

Lerp means linear interpolation between 2 values. It is commonly used to find a point some fraction of the way along a line between 2 points, e.g. to move an object gradually between those points.

It is one of the Unity functions: [Lerp function in unity]

With the given endpoints S and E of the vectors $\vec{s}$ and $\vec{e}$ calculate the vector $\vec{p}$ for different values for the fraction of the distance between S and E



What are the components of the vector $\overrightarrow{SE}$?

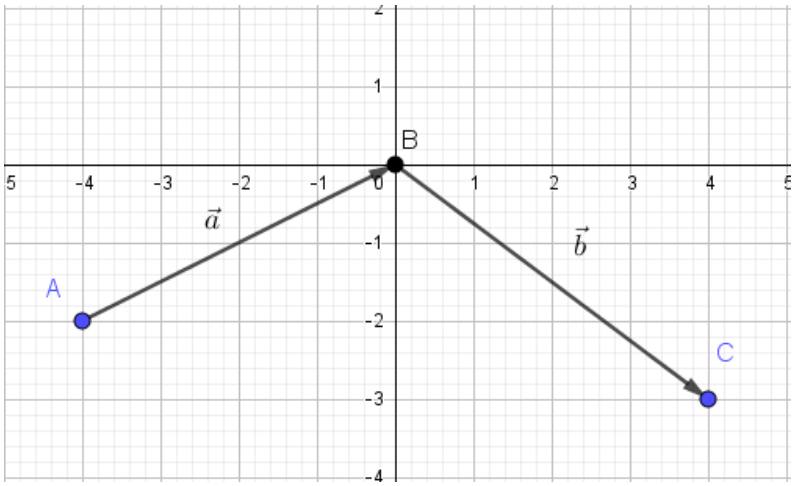
Fraction	Components of $\vec{p}$
0	
0.1	
0.2	
0.5	
0.8	
1	

Give the general formula for this vector $\vec{p}$:

1.3.5. Vertex normal

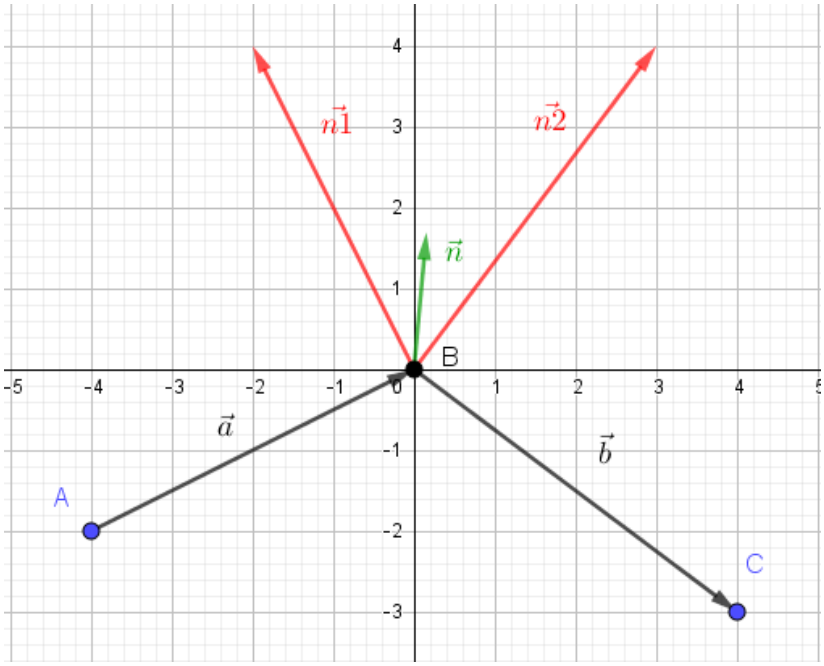
In 3D computer graphics objects are often described as a collection of vertices, edges and faces. One of the features of a vertex is its normal. A vertex normal describes the orientation of the neighbouring faces of that vertex and it is used by the lighting and shading techniques in 3D applications (Vertex normal).

Consider a vertex B with two neighbouring faces perpendicular to the (x,y)-plane as described by the vectors $\vec{a}$ and $\vec{b}$



Create this starting situation in GeoGebra and then compute the normal in the vertex B, like this:

- Create the orthogonal vectors $\vec{n}_1$ and $\vec{n}_2$ on $\vec{a}$ and $\vec{b}$ in Geogebra
- Create the vector $\vec{n}$ in vertex B, (obligatory use the formula from literature $\vec{n} = \overrightarrow{n1}/\left\|\overrightarrow{n1}\right\| + \overrightarrow{n2}/\left\|\overrightarrow{n2}\right\|$) in GeoGebra
  This is the result in GeoGebra:



- Now normalize the vector $\vec{n}$ (in Geogebra). Answer the components of this normalized vector.

2. References

2.1. Point in triangle

https://www.youtube.com/watch?v=HYAgJN3x4GA

2.2. Vertex normal

https://www.youtube.com/shorts/fzI4FfpfuSE

2.3. Vertices, faces, edges and smooth shading

https://www.mathsisfun.com/geometry/vertices-faces-edges.html

https://revisionmaths.com/gcse-maths/geometry-and-measures/3d-shapes

https://www.youtube.com/watch?v=PMgjVJogIbc

2.4. Lerp

https://docs.unity3d.com/ScriptReference/Vector3.html

https://docs.unity3d.com/ScriptReference/Vector3.LerpUnclamped.html